Master Team Project Summer 2025
UniMarkt

WWW site for buy/sell/exchange exclusively for Fulda students, staff and faculty

Team 2

Nirajan Khatri (NK)
nirajan.khatri@informatik.hs-fulda.de
Team Lead

Murtaza Muffazal Dhariwala (MMD)
murtaza-muffazal.dhariwala@informatik.hs-fulda.de
Frontend Lead

Hasara Nimashi Koralege (HNK)
hasara-nimashi.koralege@informatik.hs-fulda.de
Backend Lead

Rohan Kolagada (RK)
rohan.kolagada@informatik.hs-fulda.de
GitHub Master

Inchara Rao Konagolli Shrikanth (IRKS)
inchara-rao.konagolli-shrikanth@informatik.hs-fulda.de
Backend Developer

Muhammad Azeem Taufiq (MAT)
muhammad-azeem.taufiq@informatik.hs-fulda.de
DataBase Developer

Milestone 1
Submission on 08.05.2025

Prof. Dr. Rainer Todtenhöfer
Class CEO/CTO

Revision	**Revision Date**	**Submission Date**	**Author(s)**	**Description**
1.0 (Milestone 1)	06.05.2023	07.05.2025	NK, MMD, HNK, RK, IRKS, MAT	First version of Milestone 1, revised by team only

Table of contents

1. Executive Summary

UniMarkt is an online marketplace crafted for Hochschule Fulda's students, staff, and faculty, allowing them to buy, sell, rent, or exchange items. Unlike platforms like eBay Kleinanzeigen, UniMarkt is tailored to the university community, offering a safer, more affordable, and community-driven alternative that aligns with academic lifestyles and promotes sustainable practices.

Students and staff often face challenges finding affordable, reliable second-hand goods on public marketplaces, which are prone to overpriced listings and scams. This makes it difficult for budget-conscious users to access essential items securely, highlighting the need for a trusted, university-focused platform.

UniMarkt addresses this by offering a secure platform exclusive to verified Hochschule Fulda members, accessible via university credentials. Key features include product listings, in-app messaging, item rentals, exchange/barter options, job postings, package station integration, and verified accounts, ensuring trust, affordability, and convenience.

Developed by a dedicated Hochschule Fulda student team—led by Nirajan, with Murtaza Dariwala on frontend design, Rohan managing Github, Hasara and Ichara handling development, and Azeem overseeing the database—UniMarkt's competitive edge lies in its verified user base, on-campus job opportunities, and tailored features, making it a seamless, trusted solution for the Fulda community.

2. Personae and main Use Cases

2.1 Personae

1. **Buyer**

   A student, staff, or faculty member of Fulda University, the buyer is budget-conscious and seeks second-hand or free items from trusted, local sellers within the university community. They aim to find affordable deals conveniently and interact with reliable university-affiliated sellers. Comfortable browsing web platforms and using messaging systems, they are frustrated by wasting time on unreliable external platforms and missing out on good deals due to late discovery.

2. **Seller**

   The seller, a Fulda University student, staff, or faculty member, wants to quickly sell used items like books, electronics, or furniture to a trusted university community. They can easily upload photos and complete simple item forms but struggle to find a local audience on other platforms. They are motivated to post items and may earn free

Pro access by giving away unused items, addressing their pain point of lacking incentives for decluttering.

3. **Job Hunter**

   A Fulda University student seeking part-time or academic-related work, the job hunter wants to discover and apply for campus jobs with clear, centralized job details. They can create an academic profile including their degree, department, qualifications, LinkedIn link, and contact details. Their frustration stems from scattered job information and missing university opportunities due to poor visibility.

4. **Employer (University Staff / Faculty)**

   Faculty or staff with authority to offer part-time roles like HiWi positions, tutoring, or research support, the employer aims to post jobs simply and reach relevant student applicants quickly. Familiar with submitting job listings and descriptions, they face challenges with limited centralized tools for student job postings and slow circulation of job announcements via emails or notices.

5. **Skill Provider**

   A knowledgeable Fulda University student, staff, or faculty member, the skill provider seeks to monetize their expertise or gain teaching experience by offering tutoring or freelance services to the university community. They can teach specific topics, list services with pricing and session details, and communicate effectively. Their goal is to build a personal tutoring profile and earn positive feedback from clients.

6. **Skill Seeker**

   A Fulda University student or staff member, the skill seeker looks to learn new skills, receive tutoring, or get help with subjects like programming, design, or academic tools through affordable, trusted university-affiliated tutors. They can browse service listings, compare offerings, book appointments, and clearly communicate their needs via messaging, seeking personalized 1:1 sessions or workshops.

7. **Moderator (Admin)**

   The platform administrator, or moderator, ensures safety, content quality, and policy enforcement by approving item listings and job posts, removing inappropriate content, and restricting participation to valid Fulda University users. Experienced with moderation workflows and guided by community guidelines, they face the challenge of overwhelming manual checks as the platform scales.

2.2 Main Use Cases

1. **Buying an Item on the Platform**

A faculty member wants to buy a used bookshelf. He logs into the platform, applies a filter for “furniture,” and finds several affordable listings. He messages one seller (a student) to arrange a collection. He feels secure knowing all users are affiliated with Fulda University.

2. **Posting a Student Job by a Faculty Member**

   A Professor who needs a teaching assistant for his programming course. He goes to the Jobs section, fills out a form with the department, job title, description, qualifications, and his contact email. Once approved by the admin, it becomes visible to all student users, including those actively looking for work.

3. **Browsing Jobs as a Student**

   A student visits the platform and creates his or her Academic Profile, including details such as the "About" section, qualifications, areas of interest, LinkedIn account, and other contact information. The student then visits the platform’s Jobs section to look for a campus job. He or she can filter jobs by the department in which they are studying and can easily contact the professor using the "Apply with UniMarkt" option. The student can send a user-friendly message to the professor, expressing interest in the job and requesting them to take a look at the academic profile. She is grateful that she no longer needs to check multiple emails or notice boards.

4. **Offering a Service as a Skill Provider**

   A postgraduate student with expertise in academic writing tools (e.g., LaTeX and Overleaf) wants to offer tutoring. He goes to the Freelancing section, creates a service listing titled "Thesis Formatting Help with LaTeX," adds a price, availability slots, and short description. After admin approval, his service becomes discoverable. A few days later, he gets his first client and starts earning while helping fellow students.

5. **Booking a Tutoring Session as a Skill Seeker**

   A student is struggling with Kubernetes and wants hands-on help. She visits the Freelancing section and filters results by topic. She finds a listing for a Kubernetes tutorial posted by a computer science student. She messages the tutor, discusses timing and payment, and books a 1-hour session. She appreciates the affordable rate and the fact that the tutor understands the university's course context.

3. List of main data items and entities

This section defines the core entities, user types, and key data structures of the UniMarkt platform at a high level. Privileges are included where applicable to support user role-driven behavior.

1. **User Entities:** UserA Fulda University-affiliated user (student, staff, faculty) is classified as a Buyer, Seller, Pro User, Job Hunter, Employer, Moderator, Skill Provider, or Skill Seeker. Attributes include user_id, full_name, university email, user_roles, joined_date, optional profile_photo, and is_verified status. Privileges: Buyers browse/contact sellers, Sellers manage listings, Job Hunters apply for jobs, Employers post jobs, Moderators manage content/users, Skill Providers offer services, Skill Seekers purchase services.
2. **Marketplace Entities:** ItemAn item, listed by a Seller (e.g., books, electronics), has an item_id, title, description, category, price, condition, image_urls, posted_by, posted_date, and status. It appears in a searchable, filterable feed after admin moderation.
3. **Marketplace Entities:** MessageA message, identified by message_id, includes sender_id, receiver_id, context (item/job/skill), body, and timestamp. Private to involved users, it supports negotiations, job applications, or skill arrangements.
4. **Marketplace Entities:** Skill OfferingA paid service (e.g., tutoring), identified by skill_id, includes title, description, category, offered_by, price, availability_schedule, and status. Skill Seekers can request services, with payments handled on/off-platform and optional reviews.
5. **Job Portal Entities** Job ListingA student job (e.g., HiWi), identified by job_id, includes title, department, description, qualifications, posted_by, contact_email, posted_date, and status. Visible post-moderation, filterable by students.
6. **System Entities:** CategoryA category (e.g., Books, Academic Support), identified by category_id, includes a name and optional icon, aiding organization and navigation of items and skills.

3.1 Key Relationships

ENTITY A	**RELATIONSHIP**	**ENTITY B**	**DESCRIPTION**
User	Posts	Item	Seller creates item listings
User	Sends	Message	User Communicate via platform
User	Applies to	Job Listing	Job Hunters Contact employers
Employer (User)	Creates	Job Listing	Faculty/Staff post jobs
Moderator (User)	Modertes	Item/Job/User	Admins review and enforce rules
User	Owns	Academic Profile	Used when applying for jobs

4. Initial list of functional requirements

1. **User Registration:** Allow new users to register using their email address, ensuring the email ends with “@hs-fulda.de”.
2. **User Login and Logout:** Enable registered users to securely log in and out of the system.
3. **User Profile Management:** Allow users to view and edit their personal profile, including name, profile picture, and contact preferences.
4. **Post an Item for Sale:** Enable users to create a new listing with item title, description, category, images, and asking price.
5. **Edit/Delete Own Listings:** Allow users to modify or delete their own active listings.
6. **View Item Listings:** Provide users with the ability to browse all approved items for sale or exchange.
7. **Search Listings:** Allow users to search items by keyword, category, location, or price range.
8. **Filter and Sort Listings:** Enable users to filter listings by category or condition and sort by date or price.
9. **Item Detail Page:** Show detailed information about a selected item, including images, description, seller name (or pseudonym), and price.
10. **In-Site Messaging:** Provide a messaging system for buyers and sellers to communicate securely within the platform.
11. **Admin Item Approval:** Allow an admin to review and approve/reject items before they are visible to users.
12. **Admin User Management:** Enable the admin to view, warn, or block users who violate site policies.
13. **Mark Items as Sold or Exchanged:** Allow sellers to update the status of their item once the transaction is complete.
14. **Report Listing or User:** Enable users to report inappropriate listings or user behavior to the admin.
15. **Save Favorites / Wishlist:** Allow users to save listings to a favorites list for quick access later.
16. **Responsive Design:** Ensure the application adapts to different screen sizes (e.g., desktops and mobile browsers).
17. **Display Item Images and Media:** Support uploading and displaying of multiple images per listing, and optionally video clips.
18. **View User’s Listings:** Let users view all items currently listed by a specific seller.
19. **Category Management:** Allow admins to create, edit, or remove item categories.
20. **Display Legal Notice and Demo Text: Display the required message:** “Fulda University of Applied Sciences Software Engineering Project, Summer 2025 For Demonstration Only” on every page.
21. **Pack Station Integration:** Allow sellers to select a campus pack station as the delivery method, notify them when to drop off the item, and notify buyers when it’s ready for pickup using a secure code or QR-based retrieval system.

22. **Jobs Board Functionality:** Enable professors to post job openings (e.g., research assistant, tutoring), and allow students to browse, search, and apply for these jobs directly through the platform.
23. **Post a Service that can be booked:** Allow the users to book a service based hourly or for a particular event like tutoring, photography, tech support etc.

5. List of non-functional requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from the chosen cloud server.
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers.
3. All or selected application functions must render well on mobile devices.
4. Data shall be stored in the database on the team's deployment cloud server.
5. Full resolution free media shall be downloadable directly, and full resolution media for selling shall be obtained after contacting the seller/owner.
6. No more than 50 concurrent users shall be accessing the application at any time.
7. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users.
8. The language used shall be English (no localization needed).
9. Application shall be very easy to use and intuitive.
10. Application should follow established architecture patterns.
11. Application code and its repository shall be easy to inspect and maintain.
12. Google analytics shall be used (optional for Fulda teams).
13. No e-mail clients shall be allowed.
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security basic best practices shall be applied (as covered in the class) for main data items.
16. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today.
17. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development.
18. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0.
19. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "Fulda University of Applied Sciences Software Engineering Project, Summer 2025 For Demonstration Only" at the top of the WWW page. (Important to not confuse this with a real application).

6. Competitive analysis

Features	Ebay	kleinanzeigen	Facebook MarketPlace	SHPOCK	UniMartk
List Items	Yes	Yes	Yes	Yes	Yes
In-App Messaging	Yes	Yes	Yes	Yes	Yes
Rent	No	Yes	No	No	Yes
Job Posting By University	No	Yes	No	No	Yes
University Package Station	No	No	No	No	Yes
University Verified Users	No	No	No	No	Yes
UniSkill Bank	No	No	No	No	Yes

UniMarkt is a university-specific platform tailored for Hochschule Fulda, surpassing general marketplaces like eBay, Kleinanzeigen, and Facebook Marketplace. Unlike these, UniMarkt offers job postings for academic roles, an on-campus package station for secure transactions, and item rentals/exchanges to promote sustainability. Its UniSkill Bank enables students to trade services like tutoring or tech support, fostering collaboration. By verifying users via university credentials, UniMarkt ensures a trusted environment, delivering a comprehensive, community-focused solution that meets the unique needs of the Fulda academic community.

7. High-level system architecture and technologies used

The high-level system architecture for the Fulda University Buy/Sell/Share of Digital Media project will incorporate several key components to ensure its smooth operation. The main software components will include:

1. **Frontend Framework:** ReactJS, NextJS
2. **Backend Framework:** Python
3. **Database Management System:** MySQL
4. **APIs:** Custom APIs will be developed to enable seamless integration
5. **Cloud Server:** AWS for scalability and reliability
6. **Responsive UI Implementation:** Ensure functionality on desktop and mobile browsers

Server Configuration:

- **Host:** AWS EC2
- **Operating System:** Ubuntu Server
- **Web Server:** Nginx 1.26.1
- **Server-Side Language:** Python

Additional Technologies:

- **IDE:** Visual Studio Code
- **SASS:** 3.5.5
- **SSL Certificate:** Cert bot

8. Team and roles

Name	Role	E-Mail	GitHub Username	Frontend/Backend/Database/Cloud
Nirajan Khatri	Team Lead	nirajan.khatri@informatik.hs-fulda.de	nirajan-khatri	Frontend
Murtaza Muffazal Dhariwala	Frontend Lead	murtaza-muffazal.dhariwala@informatik.hs-fulda.de	MurtazaD1410	Frontend

Hasara Nimashi Koralege	Backend Lead	hasara-nimashi.koralege@informatik.hs-fulda.de	HNK-ENG	Backend
Rohan Kolagada	GitHub Master/Document Master	rohan.kolagada@informatik.hs-fulda.de	Rohan-Kolagada	Cloud
Inchara Rao Konagolli Shrikanth	Developer	inchara-rao.konagolli-shrikanth@informatik.hs-fulda.de	inchararaoks	Backend
Muhammad Azeem Taufiq	Developer	muhammad-azeem.taufiq@informatik.hs-fulda.de	azeemtofeek	Database

9. Checklist

- Team found a time slot to meet (online) outside of the class **DONE**
- GitHub master chosen **DONE**
- Team decided and agreed together on using the listed SW tools and deployment server **DONE**
- Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing **ON TRACK**
- Team lead ensured that all team members read the final M1 and agree/understand it before submission **DONE**
- GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.) **ON TRACK**